

Boolean Algebra (Advance Level)

A Foundation of Digital Electronics and Computing

Introduction

Boolean Algebra is a branch of mathematics developed by the English mathematician **George Boole** in 1854. Unlike ordinary algebra, which uses numbers, Boolean Algebra works with only **two possible values**:

- **1 (True, ON, High)**
- **0 (False, OFF, Low)**

Boolean Algebra forms the mathematical foundation of every modern digital system. Computers, smartphones, traffic light controllers, robots, industrial automation systems, and communication devices all make decisions using Boolean logic.

Inside a computer processor, billions of tiny electronic switches called **transistors** work together to perform Boolean operations. These transistors are combined to form **logic gates**, which execute Boolean expressions.

The main purpose of Boolean Algebra is to simplify logical expressions so that digital circuits become:

- Smaller
- Faster
- Less expensive
- More reliable
- More energy efficient

For this reason, Boolean Algebra is one of the most important topics in Digital Electronics and Computer Science.



Basic Boolean Operations

Boolean Algebra uses three fundamental operations.

1. AND Operation ($\cdot$)

The **AND** operation produces an output of **1 only when every input is 1**.

Expression: $A \cdot B$

Truth Table (AND gate)

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Circuit Symbol



Explanation

Returns **TRUE** only if both inputs are **TRUE**

TRUE AND TRUE = TRUE

FALSE AND FALSE = FALSE

TRUE AND FALSE = FALSE

Example: **If**

A = Door Closed

B = Password Correct

The door opens only if **both conditions are true**.

2. OR Operation ($+$)

The OR operation produces an output of **1 if at least one input is 1**.

Expression: $A + B$



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

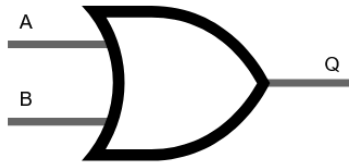
SHAWN . BRITO

shawnbritto@gmail.com

Truth Table (**OR** gate)

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Circuit Symbol



Explanation

Returns **TRUE** if either input is **TRUE**

TRUE OR FALSE = TRUE

FALSE OR FALSE = FALSE

Example: A building alarm sounds if

- Smoke is detected OR
- Fire is detected

Only one condition is required.

3. NOT Operation (–) (')

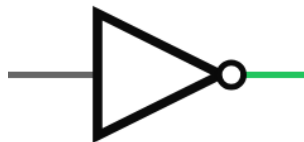
The NOT operation reverses a Boolean value.

Expression: **A'**

Truth Table

A	A'
0	1
1	0

Circuit Symbol



Explanation

Reverses the input value

NOT TRUE = FALSE

NOT FALSE = TRUE

Example: **If**

A = Light ON

Then

A' = Light OFF



Important Boolean Laws

Boolean Algebra contains many identities that simplify complex expressions.

Identity Laws

$$\mathbf{A + 0 = A}$$

$$\mathbf{A \cdot 1 = A}$$

Example

$$X + 0 = X$$

$$Y \cdot 1 = Y$$

Null Laws

$$\mathbf{A + 1 = 1}$$

$$\mathbf{A \cdot 0 = 0}$$

Example

$$P + 1 = 1$$

$$Q \cdot 0 = 0$$

Idempotent Laws

$$\mathbf{A + A = A}$$

$$\mathbf{A \cdot A = A}$$

These show that repeating the same variable has no effect.



Complement Laws

$$A + A' = 1$$

$$A \cdot A' = 0$$

Example

$$X + X' = 1$$

$$Y \cdot Y' = 0$$

Double Negation Law

$$(A')' = A$$

Taking the complement twice returns the original value.

Commutative Laws

The order of variables does not matter.

Addition

$$A + B = B + A$$

Multiplication

$$A \cdot B = B \cdot A$$

Associative Laws

Grouping does not affect the answer.

$$(A + B) + C = A + (B + C)$$

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN . BRITO
shawnbrito@gmail.com

Distributive Laws

Boolean Algebra has two distributive laws.

First Law

$$A(B + C) = AB + AC$$

Second Law

$$A + BC = (A + B)(A + C)$$

These are widely used when simplifying expressions.

Absorption Laws

The Absorption Laws help remove unnecessary terms.

Rule 1

$$A + AB = A$$

Example

$$X + XY = X$$

Because X already guarantees the output.

Rule 2

$$A(A + B) = A$$

Example

$$Y(Y + Z) = Y$$

Again, the second term contributes nothing extra. These rules significantly reduce circuit complexity.



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN . BRITO
shawnbrito@gmail.com

De Morgan's Theorems

De Morgan's Theorems are among the most important rules in Boolean Algebra. They allow us to remove brackets and convert between AND and OR operations.

First De Morgan's Theorem

$$(A + B)' = A'B'$$

Rule

The complement of an OR operation becomes an AND operation, while every variable is complemented.

Example

$$(X + Y)' = X'Y'$$

Second De Morgan's Theorem

$$(AB)' = A' + B'$$

Rule

The complement of an AND operation becomes an OR operation, while every variable is complemented.

Example

$$(XY)' = X' + Y'$$



Example Simplification

Simplify

$$(A + B)'$$

Using De Morgan's Rule

$$(A + B)' = A'B'$$

Simplify

$$(XY)'$$

Answer

$$X' + Y'$$

Why Simplification Matters

Imagine designing a processor containing billions of logic gates. If one Boolean expression can be reduced from

$$A(B + C) + AB$$

to simply

$$AB + AC$$

or even further, the processor needs:

- Fewer transistors
- Less electrical power
- Less heat generation
- Higher operating speed
- Lower manufacturing cost

This is why engineers spend considerable time simplifying Boolean expressions before building hardware.



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN . BRITO

shawnbrito@gmail.com

Summary

Boolean Algebra provides a mathematical method for representing logical decisions in digital systems.

Students should become familiar with:

- AND operation
- OR operation
- NOT operation
- Identity Laws
- Null Laws
- Complement Laws
- Distributive Laws
- Absorption Laws
- De Morgan's Theorems

Boolean Algebra

AND gate	OR gate	Absorption Rule
$A.A = A$	$A + A = A$	$A + \bar{A}.B = A + B$
$A.\bar{A} = 0$	$A + \bar{A} = 1$	$\overline{A + B} = \bar{A}.\bar{B}$
$A.1 = A$	$A + 1 = 1$	$\overline{A.B} = \bar{A} + \bar{B}$
$A.0 = 0$	$A + 0 = A$	DeMorgan's Rule

Mastering these rules allows complex Boolean expressions to be simplified into efficient digital circuits.

Practice Questions (Advanced Level)

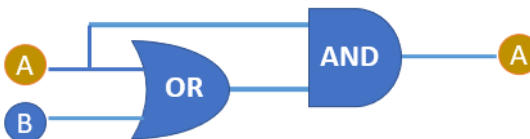
Draw the Logic-Gate diagram for the following Boolean expressions. Then simplify each Boolean expression as much as possible. For example, Question 2 has already been answered.

Question 1

$$A + AB$$

Question 2

$$A(A + B) = A$$



Question 3

$$(A + B)'$$



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN.BRITO

shawnbrito@gmail.com

Question 4

$$(AB)'$$

Question 5

$$A + A'B$$

Question 6

$$AB + AB'$$

Question 7

$$(A + B)(A + C)$$

Question 8

$$(A + B)' + AB$$

Question 9

$$(A + B)(A' + C)$$

Question 10

Simplify completely:

$$A(B + C) + A(B + C)' + AB$$





COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN . BRITO
shawnbrito@gmail.com



Challenge Question (Exam Standard)

Simplify the following expression using **Boolean Laws**, **Absorption**, and **De Morgan's Theorems**. Show every step clearly.

$$F = (A + B)' + AB + A(B + C) + A(A + C)$$

These questions progress from straightforward applications of individual laws to multi-step simplifications similar to those found in A/L ICT and Computer Science examinations.



COLOMBO (Physical) | ISLANDWIDE – ONLINE

071 436 1444

SHAWN . BRITO

shawnbrito@gmail.com